

Project Proposal: End-to-End Business Analytics Decision Platform

A Unified Data Science Solution for Enterprise Performance Monitoring, Driver Discovery, and Predictive Decision Support

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Framework:	CRISP-DM (Cross-Industry Standard Process for Data Mining)	Document ID:	DEPI-PRP-2026-V1.0
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1. Executive Summary & Background

In modern commercial enterprises, transactional data, customer interactions, partner services, and competitive market signals are generated continuously at high volume. However, organizations frequently suffer from acute *data fragmentation*: sales records reside in transactional SQL databases, marketing metrics exist behind third-party REST APIs, and external benchmarks must be scraped from web sources.

Without a consolidated analytical infrastructure, business leadership relies on fragmented, retrospective reporting that fails to explain operational drivers or forecast future trajectory. This proposal introduces the **End-to-End Business Analytics Decision Platform**, a comprehensive capstone data science initiative engineered according to the rigorous six-phase **CRISP-DM** methodology. The platform integrates heterogeneous data ingestion, automated cleaning pipelines, statistical exploratory analysis, an interactive executive dashboard, and production-ready machine learning services exposed via containerized REST APIs.

2. Problem Statement & Business Motivation

Commercial decision-makers face four critical operational bottlenecks that this project directly resolves:

- Data Silos & Inconsistency:** Enterprise operational data is disconnected across disparate systems, causing conflicting metrics, schema mismatches, and severe data reconciliation delays across business units.
- Lagging Analytical Velocity:** Manual data compilation in disconnected spreadsheets creates multi-day reporting latencies, rendering KPI monitoring retrospective rather than actionable for fast-moving market conditions.
- Absence of Predictive Foresight:** Current reporting focuses exclusively on descriptive metrics ("what happened") rather than predictive modeling ("what will happen") and prescriptive recommendations ("how to optimize").

- **Unquantified Performance Drivers:** Management lacks rigorous econometric and machine-learning attribution models to identify which operational variables drive customer churn, revenue variance, or inventory depletion.

3. Project Objectives & Measurable Success Criteria

The primary objective is to engineer a production-grade, reproducible analytics and machine-learning ecosystem that bridges raw multi-source ingestion with executive-level decision intelligence.

Objective Dimension	Specific Goal	Target Metric / KPI
Multi-Source Ingestion	Build automated ingestion connectors for SQL relational databases, external REST APIs, and public web scraping sources.	100% automated extraction; zero manual file transfers.
ETL & Governance	Implement rigorous schema validation, missing data handling, deduplication, and feature normalization pipelines.	Pipeline error rate < 0.1%; automated cleaning under 15 minutes.
Exploratory Insights (EDA)	Conduct statistical profiling, correlation discovery, cohort breakdowns, and distribution analysis across operational dimensions.	Fully documented Jupyter notebooks with actionable takeaways.
Executive Dashboard	Deploy an interactive Business Intelligence dashboard featuring KPI scorecards, trend analysis, and geographic maps.	Dashboard filter-to-render response time < 2.0 seconds.
Machine Learning Engine	Develop, benchmark, and cross-validate multiple ML models across Regression, Classification, and Clustering tasks.	Classification F1 $\geq$ 0.82; Regression R ² $\geq$ 0.80; Silhouette $\geq$ 0.50.
Production API & CI/CD	Expose model predictions via FastAPI microservice with automated GitHub Actions CI/CD and Docker containerization.	API P95 latency < 200ms; automated test coverage $\geq$ 80%.

Methodological Alignment with CRISP-DM: Each technical objective directly maps to one of the six standard CRISP-DM stages: Business Understanding establishes strategic KPIs; Data Understanding and Data Preparation establish governed data pipelines; Modeling and Evaluation implement rigorous algorithm benchmarking; and Deployment integrates models into business applications. This guarantees both academic rigor and commercial utility.

Operational Decision Velocity: By transitioning from fragmented manual data assembly to an automated, end-to-end analytics platform, organizational decision latency is reduced from days to minutes. Executive leadership gains real-time visibility into operational health while operational teams receive forward-looking predictive warnings.

4. Project Scope: In-Scope vs. Out-of-Scope

To ensure timely delivery within the DEPI timeline while maintaining enterprise quality, explicit project boundaries have been established:

In-Scope Deliverables & Functionalities	Out-of-Scope Elements & Exclusions
• Ingestion from three distinct source types: SQL DB, REST API, and Web Scraper. • Automated data cleaning, transformation, and relational feature staging. • Comprehensive statistical EDA with multivariate correlation matrices. • Executive KPI dashboard with dynamic date, segment, and geo filters. • Training, benchmarking, and hyperparameter tuning of multiple candidate ML models. • Model evaluation with confusion matrices, ROC-AUC, RMSE, and error analysis. • Containerized FastAPI inference endpoint packaged with Docker. • Automated CI/CD test suite via GitHub Actions. • Technical documentation, user manuals, and executive pitch deck.	• Sub-second real-time streaming architectures (e.g., Apache Kafka / Apache Flink). • Native mobile application development (iOS / Android). • Proprietary enterprise ERP connectors (e.g., SAP, Oracle NetSuite) requiring paid enterprise licensing. • Multi-region distributed cloud infrastructure with active-active auto-scaling. • Continuous multi-terabyte Big Data distributed processing (e.g., multi-node Spark clusters).

5. Target Stakeholders & User Personas

The platform serves distinct user cohorts across the organizational hierarchy, each with specialized interfaces and operational needs:

- **Executive Leadership (C-Suite, VP of Sales/Operations):** Consumes high-level KPI cards, revenue trajectories, forecast risk intervals, and strategic recommendations to allocate corporate budgets and track quarterly targets.
- **Business & Operations Managers:** Interacts with drill-down filters, geographic performance heatmaps, and churn risk segmentations to direct ground-level marketing and operational resources.
- **Data Analysts & BI Teams:** Leverages clean, normalized data tables, reusable SQL views, and reproducible analytical pipelines for ad-hoc querying and continuous reporting.
- **Software Engineers & Integration Partners:** Integrates with the deployed FastAPI endpoints to embed automated real-time inference within third-party operational workflows.

6. Proposed System Architecture & Technical Approach

The platform architecture is structured into four cohesive, decoupled layers:

1. **Data Acquisition Layer:** Connectors ingest transactional logs from PostgreSQL, external market indicators from third-party REST APIs (JSON/HTTP), and supplementary pricing data via automated Python web scrapers (BeautifulSoup/Selenium).
2. **Data Engineering & Transformation Layer:** Ingested raw datasets pass through automated ETL pipelines executed via Python (Pandas/NumPy) and SQL scripts, handling missing values, categorical encoding, date-time parsing, and outlier normalization. Cleaned tables are loaded into a structured analytical staging store.

3. **Analytical & Predictive Machine Learning Layer:** Statistical profiling scripts drive exploratory data analysis. The feature matrix feeds an ML experimentation pipeline (Scikit-Learn, XGBoost, LightGBM) tracked via MLflow to benchmark models, optimize hyperparameters, and register the champion model.
4. **Serving, Visualization & Presentation Layer:** Insights are surfaced through two channels: an interactive web-based BI Dashboard (Streamlit/Plotly) for analytical exploration, and a high-performance REST API (FastAPI) wrapped in a Docker container for real-time model scoring.

7. Expected Portfolio Deliverables

The final repository will showcase a complete production-grade data science portfolio containing:

- **Business Case & Problem Formulation:** Documented business context, analytical questions, and cost-benefit justification.
- **Data Pipelines & SQL Assets:** Modular extraction scripts, scraping modules, SQL schema definitions, and automated cleaning scripts.
- **Exploratory Data Analysis:** Well-documented, annotated Jupyter notebooks with statistical visualizations and executive takeaways.
- **Interactive Decision Dashboard:** Production-ready Streamlit/Plotly application with responsive controls and geographic mapping.
- **Machine Learning Suite:** Model comparison notebooks, hyperparameter logs, evaluation charts, and saved model binaries.
- **Deployment & DevOps Package:** FastAPI application source code, Dockerfile, `docker-compose.yml`, and GitHub Actions CI/CD workflows.
- **Documentation & Communication Deck:** Comprehensive README, System Architecture Document, User Manual, and Executive Presentation Slide Deck.

8. Project Feasibility & Sign-off

All proposed components utilize well-established, open-source data science frameworks with no licensing impediments. The timeline is calibrated to meet DEPI academic and industry milestone deadlines. The team is equipped with dedicated compute resources and collaborative version-control environments to deliver all deliverables by the final December 2026 evaluation.